

Adjoint Transport Theory for Rare-Event Sampling in Spin Systems

Output data appendix — all result tables, no figures

This document collects every result table produced over the course of the project, extracted from paper .tex. It contains no figures and no narrative discussion — see the paper itself for methods, interpretation, and honesty conventions (which zero-replica counts and confidence intervals mean, and which findings are established vs. suggestive). Tables are grouped by topic in the same order they appear in the paper.

1 Zero-variance demonstration and full-scale baseline

Table 1: 1D absorbing walk, exact committor available in closed form. FOM relative to naive.

Method	measured relative variance	FOM vs. naive
naive Monte Carlo	— (baseline)	$1 \times$
exact-committor tilt	$\sim 10^{-30}$ (zero to machine precision)	$\gg 1 \times$
approximate-committor tilt, pure biasing (no weight windows)	—	worse than naive

Table 2: Full scale, both models. FOM vs. WE[m] in parentheses.

Model / Method	$\hat{\pi}$	rel. s.d.	cost	FOM (vs. WE[m])	zero-rep.
<i>Ferromagnet</i> , target $ m \geq 0.9297$					
naive	9.52×10^{-6}	0.441	1.34×10^8	3.83×10^{-8} (2.92 $\times$)	0/10
WE[m]	1.05×10^{-5}	0.614	2.02×10^8	1.31×10^{-8} (1.00 $\times$)	0/10
WE[E]	1.09×10^{-5}	1.343	9.81×10^7	5.65×10^{-9} (0.43 $\times$)	2/10
WE[I_θ]	7.92×10^{-6}	0.893	1.82×10^8	6.90×10^{-9} (0.53 $\times$)	0/10
<i>EA spin glass</i> , target $E/N \leq -1.0625$					
naive	1.00×10^{-5}	0.633	1.34×10^8	1.86×10^{-8} (2.38 $\times$)	1/10
WE[m]	1.02×10^{-5}	1.857	3.72×10^7	7.80×10^{-9} (1.00 $\times$)	7/10
WE[E]	1.65×10^{-5}	1.115	1.61×10^8	5.00×10^{-9} (0.64 $\times$)	2/10
WE[I_θ]	1.54×10^{-5}	1.052	1.42×10^8	6.36×10^{-9} (0.81 $\times$)	2/10

Table 3: Incremental- R^2 diagnostic, EA spin glass, $L = 16$, matched training budget (MSE $1.0 \rightarrow 0.271$ over 20 epochs).

Quantity	value
Pearson $\text{corr}(I_\theta, E/N)$	-0.614
fraction of $\text{Var}(I_\theta)$ unexplained by E/N	0.623
$R^2(\text{label} \sim E/N \text{ alone})$	0.358
$R^2(\text{label} \sim I_\theta \text{ alone})$	0.577
$R^2(\text{label} \sim E/N + I_\theta)$	0.608
incremental R^2 from I_θ beyond E/N	+0.250

Table 4: EA spin glass, $L = 16$, one threshold step rarer than Table 2. Not a trustworthy comparison (see paper text).

Method	FOM	zero-rep.
naive	7.43×10^{-9}	4/10
WE[m]	0	10/10
WE[E]	2.18×10^{-9}	5/10
WE[I_θ]	1.56×10^{-9}	7/10

- 2 Learned coordinate: diagnostic and shallow-depth comparison
- 3 Milestoning seed sweep and deep-ensemble result
- 4 Kim & Cai self-consistency head-to-head
- 5 Deep-rare regime
- 6 Lattice-size scaling
- 7 FW-CADIS global variance reduction

Table 5: EA spin glass, $L = 16$, $T = 2.6$, one step beyond the pilot extreme, 10 replicas, milestoneing labels, seventeen independent training runs of the identical configuration. Every method observed the event in every replica in every run; naive, $\text{WE}[m]$, and $\text{WE}[E]$ are bit-identical across all runs (deterministic, and $\text{WE}[E]$ does not depend on the trained network). Seed 0 was run twice independently and reproduced bit-for-bit, confirming the network-seeding fix is complete.

Method	$\hat{\pi}$	rel. s.d.	FOM	zero-replicas
naive	5.29×10^{-5}	0.225	1.48×10^{-7}	0/10
$\text{WE}[m]$	5.55×10^{-5}	0.201	1.17×10^{-7}	0/10
$\text{WE}[E]$	4.49×10^{-5}	0.209	4.22×10^{-8}	0/10
$\text{WE}[I_\theta]$, unseeded run 1	5.03×10^{-5}	0.174	4.81×10^{-8} (+14%)	0/10
$\text{WE}[I_\theta]$, unseeded run 2	4.52×10^{-5}	0.204	3.48×10^{-8} (-17%)	0/10
$\text{WE}[I_\theta]$, seed 0 (runs 3a/3b)	5.28×10^{-5}	0.127	8.94×10^{-8} (+112%)	0/10
$\text{WE}[I_\theta]$, seed 1	5.09×10^{-5}	0.216	3.09×10^{-8} (-27%)	0/10
$\text{WE}[I_\theta]$, seed 2	5.06×10^{-5}	0.175	4.65×10^{-8} (+10%)	0/10
$\text{WE}[I_\theta]$, seed 3	4.56×10^{-5}	0.252	2.29×10^{-8} (-46%)	0/10
$\text{WE}[I_\theta]$, seed 4	4.13×10^{-5}	0.128	8.81×10^{-8} (+109%)	0/10
$\text{WE}[I_\theta]$, seed 5	5.40×10^{-5}	0.138	7.65×10^{-8} (+82%)	0/10
$\text{WE}[I_\theta]$, seed 6	4.69×10^{-5}	0.149	6.56×10^{-8} (+56%)	0/10
$\text{WE}[I_\theta]$, seed 7	4.32×10^{-5}	0.267	2.03×10^{-8} (-52%)	0/10
$\text{WE}[I_\theta]$, seed 8	4.89×10^{-5}	0.135	7.86×10^{-8} (+86%)	0/10
$\text{WE}[I_\theta]$, seed 9	5.03×10^{-5}	0.159	5.66×10^{-8} (+34%)	0/10
$\text{WE}[I_\theta]$, seed 10	4.80×10^{-5}	0.157	5.85×10^{-8} (+39%)	0/10
$\text{WE}[I_\theta]$, seed 11	4.81×10^{-5}	0.202	3.48×10^{-8} (-17%)	0/10
$\text{WE}[I_\theta]$, seed 12	5.14×10^{-5}	0.230	2.76×10^{-8} (-35%)	0/10
$\text{WE}[I_\theta]$, seed 13	4.83×10^{-5}	0.150	6.36×10^{-8} (+51%)	0/10
$\text{WE}[I_\theta]$, seed 14	4.94×10^{-5}	0.146	6.76×10^{-8} (+60%)	0/10
$\text{WE}[I_\theta]$, seed 15	4.53×10^{-5}	0.232	2.66×10^{-8} (-37%)	0/10

Table 6: EA spin glass, $L = 16$, one step rarer than Table 5, milestoneing labels, default configuration ($n_{\text{bins}} = 50$, $R = 10$).

Method	FOM	zero-rep.
naive	3.51×10^{-8}	0/10
$\text{WE}[m]$	7.73×10^{-9}	0/10
$\text{WE}[E]$	9.21×10^{-9}	0/10
$\text{WE}[I_\theta]$	8.36×10^{-9}	0/10

Table 7: Five independent $K=5$ deep-ensemble runs, identical configuration to Table 5. FOM ratio is $\text{WE}[I_\theta] / \text{WE}[E]$ on the MC-sweep cost as originally defined (see paper text for the wall-clock-corrected ratio, which excludes 1 at $0.087\times$).

Run	$\text{WE}[I_\theta]$ FOM	rel. s.d.	ratio vs. $\text{WE}[E]$
ensemble, base seed 20	1.95×10^{-8}	0.272	$0.46\times$ (-54%)
ensemble, base seed 21	2.75×10^{-8}	0.229	$0.65\times$ (-35%)
ensemble, base seed 22	3.69×10^{-8}	0.199	$0.88\times$ (-12%)
ensemble, base seed 23	3.07×10^{-8}	0.215	$0.73\times$ (-27%)
ensemble, base seed 24	3.84×10^{-8}	0.194	$0.91\times$ (-9%)

Table 8: EA spin glass, $L = 16$, shallowest calibrated target, 10 replicas. Point estimates only; see bootstrapped confidence intervals below.

Method	$\hat{\pi}$	rel. s.d.	FOM	zero-replicas
naive	3.36×10^{-5}	0.294	8.60×10^{-8}	0/10
WE[m]	3.94×10^{-5}	0.553	8.81×10^{-8}	1/10
WE[E]	3.39×10^{-5}	0.671	1.33×10^{-8}	0/10
WE[I_θ] surrogate	2.92×10^{-5}	0.904	8.62×10^{-9}	1/10
WE[I_θ] self-consistent	4.01×10^{-5}	0.655	1.62×10^{-8}	0/10

Table 9: Bootstrapped FOM ratios, full scale ($L = 16$), shallowest calibrated target, at $R = 10$ and after tripling to $R = 30$. Ratio > 1 favors the self-consistency objective.

Comparison	R	FOM ratio (median)	95% CI	significant?
self-consistent / surrogate	10	1.83	[0.49, 9.23]	no
self-consistent / surrogate	30	1.39	[0.65, 2.95]	no
self-consistent / WE[E]	10	1.19	[0.33, 4.95]	no
self-consistent / WE[E]	30	1.07	[0.55, 2.09]	no

Table 10: Same comparison as Table 9, one threshold step deeper (“beyond” = 1 instead of 0), $R = 10$.

Comparison	FOM ratio (median)	95% CI	zero-rep. (surr. / self-cons.)
self-consistent / surrogate	1.14	[0.27, 4.81]	2/10 / 1/10
self-consistent / WE[E]	1.33	[0.34, 5.74]	—

Table 11: Naive-only depth calibration, 10 replicas, custom-preset budget. Naive is fully reliable through beyond = 2, starts failing at beyond = 3, and is completely blind at beyond = 4 and beyond.

beyond	$\hat{\pi}$	rel. s.d.	zero-replicas
1	5.29×10^{-5}	0.23	0/10
2	5.48×10^{-6}	0.46	0/10
3	7.14×10^{-7}	1.61	7/10
4	0.0	—	10/10

Table 12: EA spin glass, $L = 16$, beyond = 4, self-consistency objective, three independent training seeds, 30 replicas each. WE[E]/WE[m]/naive do not depend on the trained network and are bit-identical across all three runs. Reliability: pooled 60/90 (66.7%) hits vs. WE[E]’s 12/30 (40.0%), Fisher exact $p = 0.017$ — the only statistically significant positive result for a learned coordinate in the whole project. FOM ratio (geometric-mean, two-level bootstrap over the three seeds): $1.16\times$, 95% CI [0.65, 2.12] — directionally consistent but not significant.

Method	hits/30	FOM	ratio vs. WE[E]
naive	1/30	2.48×10^{-10a}	—
WE[m]	5/30	1.04×10^{-10}	—
WE[E]	12/30	1.69×10^{-10}	1.00
WE[I_θ], seed 0	19/30	2.18×10^{-10}	$1.29\times$
WE[I_θ], seed 1	20/30	1.81×10^{-10}	$1.07\times$
WE[I_θ], seed 2	21/30	1.77×10^{-10}	$1.04\times$

^aNot a meaningful FOM: driven by a single hit in 30 replicas (rel. s.d. = 5.5); reported for completeness only.

Table 13: EA spin glass, beyond $\alpha = 1$, naive vs. the best-performing WE variant at each L (which variant wins changes with L ; see paper text). $\hat{n}$ for naive shown to indicate how rare the target is at each scale.

L	naive $\hat{n}$	naive FOM	best WE FOM (method)	ratio
8	1.79×10^{-4}	2.00×10^{-6}	1.04×10^{-6} (WE[E])	$1.9 \times$
16	5.18×10^{-5}	1.72×10^{-7}	4.83×10^{-8} (WE[m])	$3.5 \times$
24	1.30×10^{-5}	2.79×10^{-8}	8.06×10^{-9} (WE[E])	$3.5 \times$
32	4.17×10^{-6}	6.11×10^{-9}	1.69×10^{-9} (WE[I_θ])	$3.6 \times$

Table 14: FW-CADIS across every configuration tested, laptop scale through the ferromagnet control. Flatness marked * is computed only over thresholds actually observed, not comparable to unmarked rows in the same group. Dashes mark columns that do not apply to that group (coverage was 6/6 throughout past the laptop scale; a bootstrap CI was only computed for the final two configurations).

Allocation	coverage	flatness (max/min)	95% CI	cost
<i>Laptop scale, $L = 12, R = 6$, 6 thresholds</i>				
uniform ($\alpha = 0$)	6/6	283.8	—	2.01×10^7
$\alpha = 0.25$	6/6	245.2	—	7.87×10^6
$\alpha = 0.5$	5/6	65.8*	—	3.41×10^6
$\alpha = 0.75$	5/6	63.5*	—	3.29×10^6
$\alpha = 1.0$	5/6	84.8*	—	3.24×10^6
naive (reference)	6/6	463.5	—	1.61×10^7
<i>Full scale, $L = 16, R = 30$, 6 thresholds (every allocation reaches 6/6 coverage)</i>				
uniform ($\alpha = 0$)	6/6	298.0	—	1.55×10^8
$\alpha = 0.25$	6/6	377.8	—	7.89×10^7
$\alpha = 0.5$	6/6	323.9	—	2.35×10^7
$\alpha = 0.75$	6/6	342.4	—	1.98×10^7
$\alpha = 1.0$	6/6	385.5	—	1.94×10^7 ($\sim 8 \times$ cheaper)
naive (reference)	6/6	445.1	—	1.34×10^8
<i>Final EA spin-glass configuration, $L = 16, R = 90$, shallowest calibrated family</i>				
Uniform ($\alpha = 0$)	—	276.1	[224, 344]	1.59×10^8
FW-CADIS ($\alpha = 1$)	—	297.3	[234, 388]	1.99×10^7 ($\sim 8 \times$ cheaper)
<i>Ferromagnet control, $L = 16, R = 30$, shallowest calibrated family</i>				
Uniform ($\alpha = 0$)	—	100.9	[65.8, 160.3]	1.96×10^8
FW-CADIS ($\alpha = 1$)	—	287.3	[194.7, 414.8]	3.09×10^7